

Module 6:

Standard Costing & Variance Analysis
Material Cost Variance (MCV)

$$SC - AC$$

$$(SR \times SQ) - (AR \times AQ)$$

Material Price Variance

$$AQ \times (SR - AR)$$

Material Usage Variance

$$SP \times (SQ - AQ)$$

Material Mix Variance

$$SP \times (RSQ - AQ)$$

Material Yield Variance

$$SP \times (SQ - RSQ)$$

1)

Basic Calculation

Material	SQ	RSQ for actual output	SP	AQ	AP	RSQ for actual input
P	450	9000	20	10000	19	9409.09
Q	400	8000	40	8500	42	8363.63
R	250	5000	60	4500	65	5227.27
Total	1100			23000		
Std loss	100					
Std output	1000			20000		

$$MCV = SC - AC$$

Material	SQ for actual output	SP	SC	AQ	AP	AC
P	9000	20	180000	10000	19	190000
Q	8000	40	320000	8500	42	357000
R	5000	60	300000	4500	65	292500
			800000			839500

$$MCV = 800000 - 839500$$

$$= -39500 \text{ (A)}$$

$$MPV = AQ \times (SP - AP)$$

Material	SP	AP	(SP-AP)	AQ	Amnt
P	20	19	1	10000	10000
Q	40	42	-2	8500	-17000
R	60	65	-5	4500	-22500
					<u>-29500(A)</u>

$$MUV = SP \times (SQ - AQ)$$

Material	SQ	AQ	(SQ-AQ)	SP	Amnt
P	9000	10000	-1000	20	-20000
Q	8000	8500	-500	40	-20000
R	5000	4500	500	60	30000
					<u>-10000(A)</u>

$$MMV = SP \times (RSQ - AQ)$$

Material	RSQ	AQ	(RSQ-AQ)	SP	Amnt(2)
P	9409.09	10000	-590.91	20	-11818.2
Q	8363.69	8500	-136.31	40	-5452.4
R	5227.27	4500	727.27	60	43636.2
					<u>26365.6(F)</u>

$$MYV = SP \times (SQ - RSQ)$$

Material	SQ	RSQ	(SQ-RSQ)	SP	Amnt
P	9000	9409.09	-409.09	20	-8181.8
Q	8000	8363.69	-363.69	40	-14547.6
R	5000	5227.27	-227.27	60	-13636.2
					<u>-36365.6(A)</u>

Reconciliation

$$MCV = MPV + MUV$$

$$39500(A) = 29500 + 10000$$

$$39500(A) = 39500$$

$$MOV = MMV + MYV$$

$$10000(A) = 26365.6 + 36365.6$$

$$10000(A) = 10000$$

2)

Materials	SM(%)	SQ	SQ per std output	SP	AM(%)	AQ	AP	input RSQ per Actual
X	50	62.5	3500	40	60	4200	42	3500
Y	30	37.5	2100	20	20	1400	16	2100
Z	20	25	1400	10	20	1400	12	3500
Total		125				7000		1400
Less								
Output		100				5600		

$$MCV = SC - AC$$

Materials	SQ per 5600kg	SP	SC	AQ	AP	AC
X	3500	40	140000	4200	42	176400
Y	2100	20	42000	1400	16	22400
Z	1400	10	14000	1400	12	16800
			16000			215600

$$= -19600(A)$$

$$MPV = AQ \times (SP - AP)$$

Materials	SP	AQ	(SP - AP)	AQ	Amnt(₹)
X	40	42	2	4200	-8400
Y	20	16	4	1400	5600
Z	10	12	-2	1400	-2800

$$MW = SP \times (SQ - AQ)$$

Materials	SQ	AQ	(SQ - AQ)	SP	Amnt(₹)
X	3500	4200	-700	40	-28000
Y	2100	1400	700	20	14000
Z	1400	1400	0	10	0
					-14000(A)

$$MMV = SP \times (RSQ - AQ)$$

Materials	RSQ	AQ	(RSQ - AQ)	SP	Amnt(₹)
X	3500	4200	-700	40	-28000
Y	2100	1400	700	20	14000
Z	1400	1400	0	10	0
					-14000(A)

$$MYV = SP \times (SQ - RSQ)$$

Materials	SQ	RSQ	(SQ - RSQ)	SP	Amt(₹)
X	3500	3500	0	40	0
Y	2100	2100	0	20	0
Z	1400	1400	0	10	0
					<u>0</u>

Reconciliation:

$$MCV = MPV + MUW$$

$$19600(A) = 5600(A) + 14000(A)$$

$$MUW = MMU + MYV$$

$$14000(A) = 14000(A) + 0$$

3) Basic calculation

Materials	SM(%)	SQ	^{Standard} Output	SP	AQ	AP	RSQ (P) actual input
A	40	160	161.78	20	180	18	160
B	60	240	242.67	30	220	34	240
Total	100	400			2400		
Loss	10	40					
Output	90	360					

$$MCV = SC - AC$$

Material	SQ	SP	SC	AQ	AP	AC
A	161.78	20	3235.56	180	18	3240
B	242.67	30	7280	220	34	7480
			10515.56			10720
						= -204.44(A)

$$MPV = AQ \times (SP - AP)$$

Material	SP	AP	(SP - AP)	AQ	Amt(₹)
A	20	18	2	180	360
B	30	34	-4	220	-880
					<u>520(A)</u>

$$MUU = SP \times (ASQ - AQ)$$

Materials	SP	AQ	(SQ-AQ)	SP	Amnt(₹)
A	161.78	180	-18.22	20	-364.44
B	242.67	220	22.67	30	680
					<u>315.56(F)</u>

$$MMU = SP \times (RSQ - AQ)$$

Materials	RSQ	AQ	(RSQ-AQ)	SP	Amnt(₹)
A	160	180	-20	20	-400
B	240	220	20	30	600
					<u>200(F)</u>

$$MYU = SP \times (SQ - RSQ)$$

Materials	SP	RSQ	(SQ-RSQ)	SP	Amnt(₹)
A	161.78	160	1.78	20	35.56
B	242.67	240	2.67	30	80
					<u>115.56(F)</u>

Reconciliation

$$MCU = MPV + MUU$$

$$204.44(A) = 520(A) + 315.56(F)$$

$$MUU = MMU + MYU$$

$$315.56(F) = 200(F) + 115.56(F)$$

4)

Basic calculation

Materials	SP(₹)	SO	SO per actual output	SP	AQ	AP	RSQ for AC input
A	75	2250	2375	2	2200	2.11	2250
B	25	750	791.67	10	800	9.81	750
total	100	3000			3000		
loss	10	300					
output	90	2700					

$$MCU = SC - AC$$

Material = SC - AC

Material	SQ	SP	SC	AQ	AP	AC
A	2375	2	4750	2200	2.11	4650
B	791.67	10	7916.67	800	9.81	7848
			12666.67			12498

$$= 168.67(F)$$

MPV = AQ x (SP - AP)

Material	SP	AP	(SP - AP)	AQ	Amnt(₹)
A	2	2.11	-0.11	2200	-250
B	10	9.81	0.19	800	152
					-98(A)

MCU = SP x (SQ - AQ)

Material	SQ	AQ	(SQ - AQ)	SP	Amnt(₹)
A	2375	2200	175	2	350
B	791.67	800	-8.33	10	-83.33
					266.67(F)

MMV = SP x (RSQ - AQ)

Material	RSQ	AQ	(RSQ - AQ)	SP	Amnt(₹)
A	2250	2200	50	2	100
B	750	800	-50	10	-500
					-400(A)

MYU = SP x (SQ - RSQ)

Material	SQ	RSQ	(SQ - RSQ)	SP	Amnt(₹)
A	2375	2250	125	2	250
B	791.67	750	41.67	10	416.67
					666.67(F)

Reconciliation

$$MCU = MPV + MMV$$

$$168.67(F) = 98(A) + 266.67(F)$$

$$MN = MMU + MN$$

$$266.67(F) = 400(A) + 666.67(F)$$

WN:

400 1000 for SH
F 960 Actual

Basic calculation

Composition	SM	Hours	SH	Super A output	SR	AM	AH	AR
Men	10	40	400		0.625	13		0.6
Women	5	40	200		0.4	4		0.425
Boys	5	40	200		0.35	3		0.325
Total	20					20		
Idle time		2						
Std output			1000				960	

Std Input is the same as actual input for RSH for actual input

$$LCU = SH \times SR - (AH \times AR) - (SC - AC)$$

Composition	SH	SR	SC	AH	AR	AC
Men	384	0.625	240	520	0.6	312
Women	192	0.4	768	160	0.425	68
Boys	192	0.35	67.2	120	0.325	39
			384			419

$$= -35(A)$$

$$LRV = AH \times (SR - AR)$$

Composition	SR	AR	(SR - AR)	AH	Amount(£)
Men	0.625	0.6	0.025	520	13
Women	0.4	0.425	(0.025)	160	(4)
Boys	0.35	0.325	0.025	120	3
					12(F)

$$LFU = SR \times (SH - AH)$$

Composition	SH	AH	(SH - AH)	SR	Amount(£)
Men	384	520	(136)	0.625	(85)
Women	192	160	32	0.4	12.8
Boys	192	120	72	0.35	25.2
					47(A)

$$LMV = SR \times (RSH \text{ for actual input} - AH)$$

Composition	RSH	AH	(RSH-AH)	SR	Amnt(₹)
Men	400	520	(120)	0.625	(75)
Women	200	160	40	0.4	16
Boys	200	120	80	0.35	28
					31(A)

$$LYV = SR \times (SH - RSH \text{ for hrs utilized})$$

Composition	SH	RSH for	(SH-RSH)	SR	Amnt(₹)
Men	384	380	4	0.625	2.5
Women	192	190	2	0.4	0.8
Boys	192	190	2	0.35	0.7
					4(F)

$$LITV = SM \times SR \times \text{Idle hrs}$$

Composition	Idle hrs	SM	SR	Amnt(₹)
Men	2	10	0.625	12.5
Women	2	5	0.4	4
Boys	2	5	0.35	3.5
				20(A)

Though positive it is an adverse situation.
Idle time represents a loss/wastage

Reconciliation:

$$LCV = LRV + LEV$$

$$35(A) = 2(F) + 47(A)$$

$$LEV = LMV + LYV + LITV$$

$$47(A) = 31(A) + 4(F) + 20(A)$$

9)

Composition	SM	Hours	SH	SH per output	SR	AM	AR	AR	RSH per actual input
Skilled	2	200	400	405	20	2	400	20	400
Semi-skilled	4	200	800	810	12	3	600	14	800
Unskilled	4	200	800	810	8	5	1000	10	800
	<u>10</u>		<u>2000</u>			<u>10</u>	<u>2000</u>		
Total		12							
Idle time standard output			800				810		

RSH per
hours utilized
(188 hours)

376

752

752

$$LEV = (SH \times SR) - (AM \times AR) = SC - AC$$

Composition	SH per AO	SR	SC	AM	AR	AC
Skilled	405	20	8100	400	20	8000
Semi-skilled	810	12	9720	600	14	8400
Unskilled	810	8	6480	1000	10	10000
			<u>24300</u>			<u>26400</u>

$$= -2100(A)$$

$$LRV = AM \times (SR - AR)$$

Composite	SR	AR	(SR-AR)	AM	Amnt(2)
Skilled	20	20	0	400	0
Semi-skilled	12	14	-2	600	-1200
Unskilled	8	10	-2	1000	-2000
					<u>-3200(A)</u>

$$LEV = SR \times (SH - AH)$$

Composition	SH for AD	AH	(SH-AH)	SR	Amnt(₹)
Skilled	405	400	5	20	100
Semi-skilled	810	600	210	12	2520
Unskilled	810	1000	-190	8	-1520
					<u>1100(F)</u>

$$LMV = SR \times (RSH \text{ for actual input} - AH)$$

Composition	RSH	AH	(RSH-AH)	SR	Amnt(₹)
Skilled	400	400	0	20	0
Semi-skilled	800	600	200	12	2400
Unskilled	800	1000	-200	8	-1600
					<u>800(F)</u>

$$LYV = SR \times (SH - RSH \text{ for actual hours utilized})$$

Composition	SH	RSH for hours utilized	SH-RSH	SR	Amnt(₹)
Skilled	405	376	29	20	580
Semi-skilled	810	752	58	12	696
Unskilled	810	752	58	8	464
					<u>1740(F)</u>

$$LITV = SM \times SR \times \text{Idle hours}$$

Composition	Idle hours	SM	SR	Amnt(₹)
Skilled	12	2	20	480
Semi-skilled	12	4	12	576
Unskilled	12	4	8	384
				<u>1440(A)</u>

Reconciliation

$$LCV = LRV + LEV$$

$$\text{₹ } 2100(A) = 3200(A) + 1100(F)$$

$$LEV = LMV + LYV + LITV$$

$$1100(F) = 800(F) + 1740(F) + 1440(A)$$

Basic calculation

Composition	SM	Hours	SM	SM for A output	SR	PM	PM	AR	RSN for Actual in	Revised SM
Skilled	30	2000	6000	5760	5	24	4800	6	6120	30.6
Semiskilled	10	200	2000	1920	3	15	3000	2.5	2040	10.2
Unskilled	10	200	2000	1920	2	12	2400	2	2040	10.2
Total			10000				10200			51
Idle time standard output		15		10000			9600			

RSN for (SM x hours utilized)
(185 hours)
5661
1887
1887

$$LCU = (SM \times SR) - (PM \times AR) = SC - AC$$

Composition	SM for AO	SR	SC	PM	AR	AC
Skilled	5760	5	28800	4800	6	28800
Semiskilled	1920	3	5760	3000	2.5	7500
Unskilled	1920	2	3840	2400	2	4800
			38400			41100

$$= -2700(A)$$

$$LRU = PM \times (SR - AR)$$

Composition	SR	AR	(SR - AR)	PM	LRU (A)
Skilled	5	6	-1	4800	-4800
Semiskilled	3	2.5	0.5	3000	1500
Unskilled	2	2	0	2400	0
					-3300(A)

$$LEV = SR \times (SH - AM)$$

Composition	SH	AM	(SH-AM)	SR	Amnt(₹)
Skilled	5760	4800	960	5	4800
Semi-skilled	1920	3000	-1080	3	-3240
Unskilled	1920	2400	-480	2	-960
					<u>600(₹)</u>

$$LMV = SR \times (RSH \text{ from actual input} - AM)$$

Composition	RSH	AM	(RSH-AM)	SR	Amnt(₹)
Skilled	6120	4800	1320	5	6600
Semi-skilled	2040	3000	-960	3	-2880
Unskilled	2040	2400	-360	2	-720
					<u>3000(₹)</u>

$$LYU = SR \times SH - (RSH \text{ for actual hours utilized})$$

Composition	SH	RSH for Actual hours	SH - SRSH	SR	Amnt(₹)
Skilled	5760	5661	99	5	495
Semi-skilled	1920	1887	33	3	99
Unskilled	1920	1887	33	2	66
					<u>660(₹)</u>

$$LTV = SM \times SR \times \text{Idle hours}$$

Composition	Idle hours	SM	SR	Amnt(₹)
Skilled	15	30.6	5	2295
Semi-skilled	15	10.2	3	459
Unskilled	15	10.2	2	306
				<u>3060(₹)</u>

Per Utilization

$$LCU = LRV + LEV$$

$$2700(A) = 3300(A) + 600(F)$$

$$600 = 12000$$

classmate

Date

Page

$$LEV = LMV + LYV + LITV$$

$$600(F) = 3000(F) + 660(F) + 3060(F)$$

11)

Basic calculation

Particulars	Standard	Actual
Output	600	500
Standard Time for 1 unit	20	
Total hrs	12000 hrs	9000 hrs
Overhead cost	15600	14000
Overhead rate	1.3	1.56
Standard hrs for Actual output	10000	

$$\begin{aligned} VOCV &= (SH \times SR) - (AH \times AR) = SC - AC \\ &= (10000 \times 1.3) - (9000 \times 1.56) \\ &= 13000 - 14040 \\ &= -1040(A) \end{aligned}$$

$$\begin{aligned} VOEV &= AH \times (SR - AR) \\ &= 9000 \times (1.3 - 1.56) \\ &= 9000 \times -0.26 \\ &= -2340(A) \end{aligned}$$

$$\begin{aligned} VOEFV &= SR \times (SH - AH) \\ &= 1.3 \times (10000 - 9000) \\ &= 1.3 \times 1000 \\ &= 1300(F) \end{aligned}$$

Reconciliation

$$\begin{aligned} VOCV &= VOEV + VOEFV \\ -1040(A) &= -2340(A) + 1300(F) \end{aligned}$$

12)

Basic computation

Particulars	Standard	Actual
Output	300	250
No of hrs	20	-
Total No of hrs	6000 hrs	4500 hrs
Overhead cost	7800	7000
Overhead rate	1.3	1.56
Standard hrs for actual output	5000	

$$\begin{aligned}
 \text{VOCV} &= (\text{SH} \times \text{SR}) - (\text{AH} \times \text{AR}) \\
 &= (5000 \times 1.3) - (4500 \times 1.56) \\
 &= 6500 - 7020 \\
 &= -520 (\text{A})
 \end{aligned}$$

$$\begin{aligned}
 \text{VOEU} &= \text{AH} \times (\text{SR} - \text{AR}) \\
 &= 4500 \times (1.3 - 1.56) \\
 &= 4500 \times -0.26 \\
 &= -1170 (\text{A})
 \end{aligned}$$

$$\begin{aligned}
 \text{VOEFV} &= \text{SR} \times (\text{SH} - \text{AH}) \\
 &= 1.3 \times (5000 - 4500) \\
 &= 1.3 \times 500 \\
 &= 650 (\text{F})
 \end{aligned}$$

Reconciliation

$$\text{VOCV} = \text{VOEU} + \text{VOEFV}$$

$$-520 (\text{A}) = -1170 (\text{A}) + 650 (\text{F})$$

$$\begin{aligned}
 \text{Variable overhead Idle time variance} &= \text{Idle hours} \times \text{SR} \\
 &= 300 \times 1.3 \\
 &= 390 (\text{A})
 \end{aligned}$$

classmate
Date _____
Page _____

Variable overhead Net efficiency variance
 $= \text{Basic efficiency} - \text{Idle time}$
 $= 650(F) - 390(A)$
 $= 1040(F)$

Reconciliation:

$$VOCV = VOEV + VOV + VOITV$$

$$520(A) = 1170(A) + 1040(F) + 390(A)$$

Basic computation

Particulars	Standard	Actual
	400	360
Output	-	-
No of hrs	8000	7000
Total No of hrs	10000	9150
Overhead cost	1.25	1.31
Overhead rate	7200	
SH for A output		

$$VOCV = (SH \times SR) - (AH \times AR)$$

$$= (7200 \times 1.25) - (7000 \times 1.31)$$

$$= 9000 - 9170$$

$$= -170(A)$$

$$VOEV = AH \times (R - AR)$$

$$= 7000 \times (1.25 - 1.31)$$

$$= 7000 \times -0.06$$

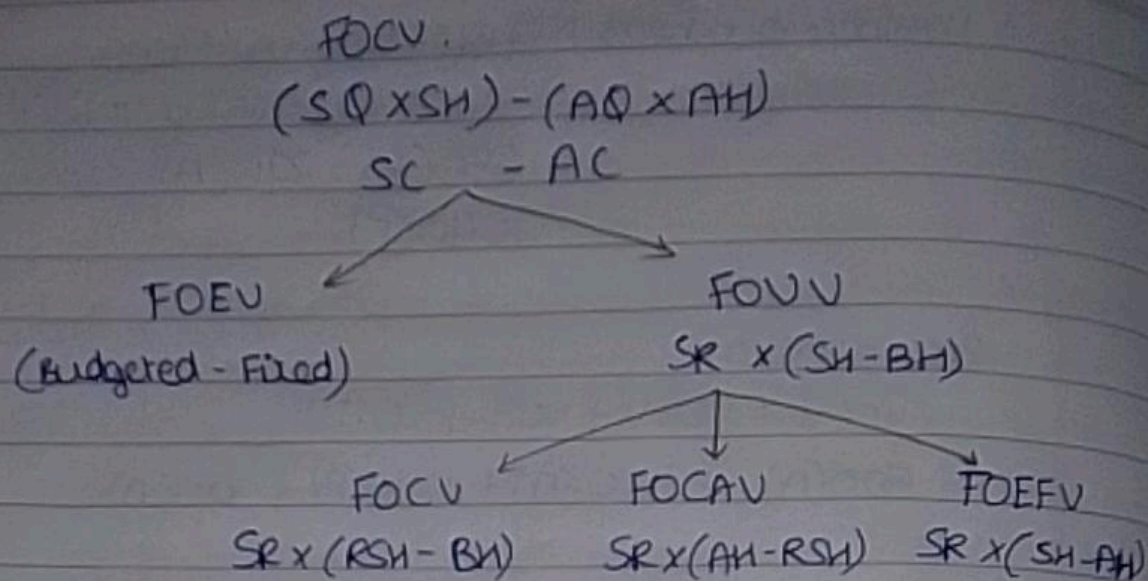
$$= -420(A)$$

$$VOEFV = SR \times (SH - AH)$$

$$= 1.25 \times (7200 - 7000)$$

$$= 1.25 \times 200$$

$$= 250(F)$$



4.)

Particulars	Standard	Actual
No of working days	20	22
Man hours per day	8000	8400
Total hours	160000	184800
Output per man hour in units	1	0.9
Total output	160000	166320
Overhead cost	160000	168000
Overhead rate	1	0.91
Standard hrs for actual output	166320	-
Revised standard hrs	176000	

$$\begin{aligned} \text{FOCV} &= (SH \times SR) - (AH \times AR) = SC - AC \\ &= (166320 \times 1) - (184800 \times 0.91) \\ &= -1848 (A) \end{aligned}$$

$$\begin{aligned} \text{FOEV} &= \text{Budgeted FOH} - \text{Actual FOH} \\ &= 160000 - 168000 \\ &= -8000 (A) \end{aligned}$$

$$\begin{aligned} \text{FOUV} &= SR \times (SH - BH) \\ &= 1 \times (166320 - 160000) \\ &= 6320 (F) \end{aligned}$$

* 20000 → 5000

? → 5200

classmate

Date
Page

$$\begin{aligned} \text{FOCV} &= \text{SR} \times (\text{RSH} - \text{BH}) \\ &= 1 \times (176000 - 160000) \\ &= 1 \times 16000 \\ &= 16000(\text{F}) \end{aligned}$$

$$\begin{aligned} \text{FOCV} &= \text{SR} \times (\text{AH} - \text{RBH}) \\ &= 1 \times (184800 - 176000) \\ &= 8800(\text{F}) \end{aligned}$$

$$\begin{aligned} \text{FOEFV} &= \text{SR} \times (\text{SH} - \text{AH}) \\ &= 1 \times (166320 - 184800) \\ &= 18480(\text{A}) \end{aligned}$$

Reconciliation

$$\begin{aligned} \text{FOCV} &= \text{FOEV} + \text{FOVV} \\ 18480(\text{A}) &= 8000(\text{A}) + 6320(\text{F}) \end{aligned}$$

$$\begin{aligned} \text{FOVV} &= \text{FOCV} + \text{FOCV} + \text{FOEFV} \\ 6320(\text{F}) &= 16000(\text{F}) + 8800(\text{F}) + 18480(\text{A}) \end{aligned}$$

15) Particulars	Budgeted	Actual
Fixed overhead	10000	10200
Units of output	5000	5200
Std time per unit	4 hrs	-
Total hrs	20000 hrs	20100 hrs
Overhead rate	0.5	0.51
Standard hrs for AO	20800	

$$\begin{aligned} \text{FOCV} &= (\text{SH} \times \text{SR}) - (\text{AH} \times \text{AR}) = \text{SC} - \text{AC} \\ &= (20800 \times 0.5) - (20100 \times 0.51) \\ &= 149(\text{F}) \end{aligned}$$

$$\begin{aligned} \text{FOEV} &= \text{Budgeted FOH} - \text{Actual FOH} = 10000 - 10200 \\ &= 200(\text{A}) \end{aligned}$$

$$\begin{aligned} FOUV &= SR \times (SH - BH) \\ &= 0.5 \times (20800 - 20000) \\ &= 400(F) \end{aligned}$$

$$\begin{aligned} FOCV &= SR \times (RSH - BH) \\ &= 0.5 \times (20000 - 20000) \\ &= 0 \end{aligned}$$

$$\begin{aligned} FOCapV &= SR \times (AM - RBH) \\ &= 0.5 \times (20100 - 20000) \\ &= 50(F) \end{aligned}$$

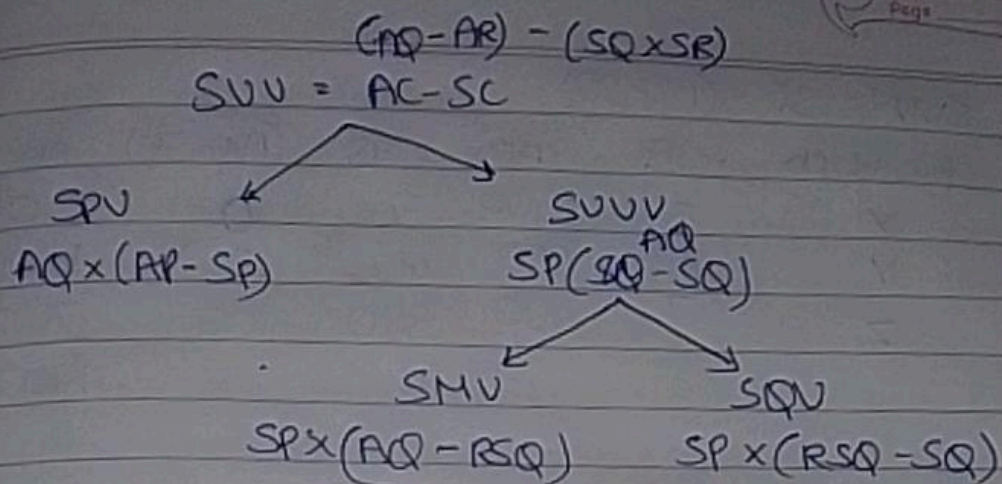
$$\begin{aligned} FOEFV &= SR \times (SH - AM) \\ &= 0.5 \times (20800 - 20100) \\ &= 350(F) \end{aligned}$$

Reconciliation:

$$\begin{aligned} FOCV &= FOEV + FOUV \\ 149(F) &= 200(A) + 400(F) \end{aligned}$$

$$\begin{aligned} FOUV &= FOCV + FOCapV + FOEFV \\ 400(F) &= 0 + 50(F) + 350(F) \end{aligned}$$

P.T.O. →



16)

Basic calculation

Products	AQ	AP	SQ	SP	RSQ
X	500	5	500	5	625
Y	600	6.25	400	6	500
Z	400	6.75	300	7	375
	<u>1500</u>		<u>1200</u>		

$$500 = 1200$$

$$7 = 1500$$

$$SVV = AC - SC = (AQ \times AP) - (SQ \times SP)$$

Materials	AC	SC	AC-SC
X	2500	2500	0
Y	3750	2400	1350
Z	2700	2100	600
			<u>1950 (F)</u>

$$SPV = AQ \times (AP - SP)$$

Products	AQ	AP	SP	(AP-SP)	amt(z)
X	500	5	5	0	0
Y	600	6.25	6	0.25	150
Z	400	6.75	7	-0.25	-100
					<u>50 (F)</u>

$$SUM = SP \times (AQ - SQ)$$

Products	SP	AQ	SQ	(AQ-SQ)	Amt(z)
X	5	500	500	0	0
Y	6	600	400	200	1200
Z	7	400	300	100	700
					<u>1900(F)</u>

$$SMU = SP \times (AQ - RSQ)$$

Products	SP	AQ	RSQ	AQ-RSQ	Amt(z)
X	5	500	625	-125	-625
Y	6	600	500	100	600
Z	7	400	375	25	175
					<u>150(F)</u>

$$SQV = SP \times (RSQ - SQ)$$

Products	SP	RSQ	SQ	RSQ-SQ	Amt(z)
X	5	625	500	125	625
Y	6	500	400	100	600
Z	7	375	300	75	525
					<u>1750(F)</u>

Reconciliation

$$SUV = SPV + SUVV$$

$$1950(F) = 50(F) + 1900(F)$$

$$SUVVV = SMU + SQV$$

$$1900(F) = 150(F) + 1750(F)$$